Machine Learning-Based Prediction of Marketing Campaign Revenue and Product Performance Using Supervised Learning

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Abstract

This work introduces a supervised learning model for estimating the revenue from a marketing campaign using regression algorithms in machine learning. The experiment was conducted using the Marketing and Product Performance dataset provided by Kaggle. Several regression algorithms such as Linear Regression, Decision Tree, K-Nearest Neighbor, Random Forest, and XGBoost were tested using the MAE, RMSE, and R^2 criteria. The experimental results showed that SVR achieved the least negative R^2 score, while ensemble models were also evaluated for nonlinear revenue prediction.

Index Terms

machine learning, regression, marketing analytics, revenue prediction, supervised learning

I. Introduction

One of the areas of research gaining more importance in recent times is marketing analytics owing to increased availability of marketing and campaign data in the digital business world. Businesses collect huge amounts of marketing data through their advertising campaigns, online sale systems, customer subscriptions, and various marketing initiatives. Data analysis could assist organizations in enhancing their decision-making process, making their marketing efforts more effective, and boosting revenues. Machine learning algorithms have been widely popular among researchers owing to their ability to find patterns and correlations within big data sets. One of the best ways to make predictions is regression-based predictive modeling. Machine learning techniques have gained significant attention in predictive analytics and business intelligence applications [1], [2].

Revenue prediction becomes essential in assisting firms with budget allocation and the assessment of marketing campaigns' success. Conventional methods of statistics may fail to model nonlinear relationships that exist between many marketing attributes and the generation of revenue. The application of supervised learning algorithms represents a more adaptive method of dealing with complicated datasets within the business setting. Over the past few years, machine learning algorithms like Decision Trees, Random Forests, Support Vector Machines, and Gradient Boosting techniques have proven successful in business intelligence applications.

A. Related Work

A number of works have examined machine learning methods used in marketing analytics, sales prediction, and business intelligence systems. Regression machine learning algorithms have gained popularity due to their ability to predict customers' behavior, product demand, and revenues using historical information about business performance. Traditionally, linear regression is used as a common approach for performing forecasts; nevertheless, more recent literature proves that ensemble learning techniques and machine learning algorithms that do not rely on linearity produce higher prediction power when dealing with sophisticated data sets. Random Forest and Gradient Boosting algorithms proved to be highly effective at processing nonlinear relationships among features present in marketing datasets. Prior research has shown how preprocessing, feature engineering, and model tuning affect the accuracy of regression algorithms. Regression-based machine learning models are widely used for predictive analytics and forecasting applications [1]–[5].

B. Dataset

The dataset used in this study is the Marketing and Product Performance Dataset obtained from Kaggle [6]. This data set has a total number of 10,000 rows and 17 variables related to marketing campaigns, subscriptions, sales, and earning revenue. Some of the key variables in this data set include Budget, Clicks, Conversions, ROI, Subscription Tier, Discount, Units Sold, Bundle

Cost, and Post-refund Customer Satisfaction. The target variable that will be predicted in this case is Revenue Generated. The data set was first cleaned through the elimination of duplicates, the removal of irrelevant columns, filling of missing values, and the use of label encoding methods on the categorical variables.

C. Gap Analysis

Despite a number of machine learning methods being used for analyzing marketing data and predicting sales, the existing literature mostly considers classification problems or involves small-size datasets that contain few variables related to business. Currently, there is no empirical work comparing the performance of regression techniques for predicting marketing revenue against that of more complex models, such as classification algorithms and ensemble learners. Another gap in the literature relates to evaluating the effects of data preprocessing and assessing the robustness of predictions in terms of noise in data. There is no prior research that examines features' contributions to predictions in marketing campaigns.

D. Problem Statement

Marketing organizations generate large volumes of campaign and product performance data. Predicting revenue generated from marketing activities is important for optimizing campaign effectiveness, budgeting strategies, and business decision-making. Traditional statistical approaches often struggle to capture complex relationships between marketing variables and revenue generation. Therefore, this study applies supervised learning regression techniques to predict marketing revenue using campaign performance indicators and customer-related attributes.

1) How effectively can baseline supervised learning regression models predict marketing revenue generation?
2) Which supervised learning regression model achieves the highest predictive accuracy?
3) How do preprocessing techniques influence regression model performance?
4) Which marketing and product performance features contribute most significantly to revenue prediction?
5) How robust are the selected regression models under different validation and noisy data conditions?

E. Novelty of our work and Our Contributions

This paper proposes a full-fledged machine learning based marketing revenue prediction methodology. Whereas most studies rely solely on comparing one model at a time, this paper focuses on the comparative performance analysis of baseline and advanced regression-based machine learning models such as Linear Regression, Decision Tree, K-Nearest Neighbors, Random Forest, Support Vector Regression, and XGBoost. Preprocessing analysis, importance of features, and robustness under various conditions have been analyzed.

This paper makes several major contributions in terms of comparative analysis of different regression-based models for predicting marketing revenue, the effect of preprocessing, analysis of important features, and robustness analysis through noise and cross validation conditions. From the experimental results, advanced regression models provided a broader comparison against baseline models, with SVR achieving the strongest R^2 result among the evaluated models.

II. Methodology

A. Overall Workflow

The workflow of the proposed approach can be described in terms of several steps: collecting the data, performing pre-processing, carrying out feature extraction, training machine learning algorithms, evaluating their performance, and analyzing the results obtained. In the first step of the approach, the dataset containing information about the Marketing and Product Performance was downloaded from Kaggle and checked for missing values, duplicates, and irrelevant features. Unnecessary identifier columns and categorical variables were eliminated, and encoding Label Encoding technique was applied to all categorical variables. Then, the dataset was split into train and test datasets in a ratio of 80:20. Multiple supervised learning regression models including Linear Regression, Decision Tree, K-Nearest Neighbors, Random Forest, Support Vector Regression, and XGBoost [1], [2], [7] were trained. The performance of models was measured based on such metrics as Mean Absolute Error, Root Mean Square Error, and coefficient of determination.

III. Results

The baseline regression models were evaluated to analyze their ability to predict marketing revenue using campaign and customer-related features. Linear Regression achieved the least negative R^2 score among the baseline approaches, while Decision Tree and K-Nearest Neighbors produced higher prediction errors and lower generalization capability. These results indicate that simple baseline models struggle to capture the nonlinear relationships present within the dataset.

TABLE I
BASELINE REGRESSION MODEL PERFORMANCE

Model	MAE	RMSE	R^2
Linear Regression	24756.22	28601.12	-0.00318
Decision Tree	32103.14	39660.07	-0.92895
KNN	26306.89	31259.19	-0.19831

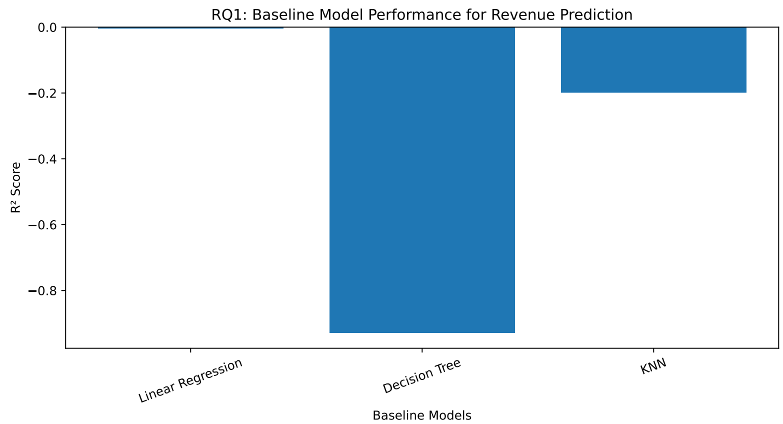

Fig. 1. Baseline regression model performance comparison using R^2 scores for revenue prediction. Linear Regression, Decision Tree, and KNN models were evaluated to establish baseline predictive performance.

The advanced regression models, namely Random Forest, Support Vector Regression, and XGBoost, were evaluated to compare predictive performance with baseline models. The results indicate that SVR achieved the least negative R^2 score, while Random Forest and XGBoost provided additional comparison for nonlinear and ensemble-based prediction.

TABLE II
COMPARATIVE REGRESSION MODEL PERFORMANCE

Model	MAE	RMSE	R^2
Linear Regression	24756.22	28601.12	-0.00318
Decision Tree	32103.14	39660.07	-0.92895
Random Forest	25138.49	29181.17	-0.04429
XGBoost	26170.23	30789.65	-0.16258
SVR	24728.57	28589.68	-0.00238

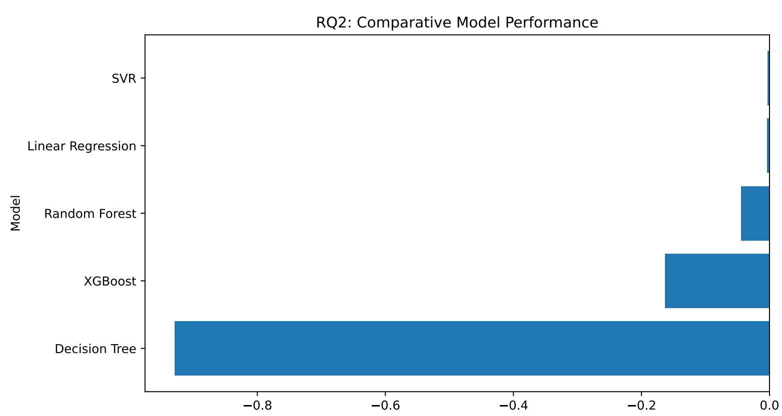

Fig. 2. Comparative regression model performance analysis using R^2 scores. The figure compares baseline and advanced models including Random Forest, SVR, and XGBoost for revenue prediction.

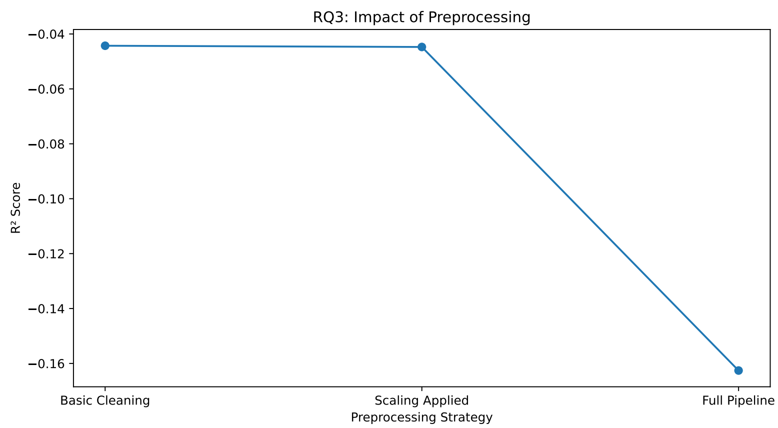

Fig. 3. Impact of preprocessing strategies on regression model performance. The figure compares basic cleaning, feature scaling, and full preprocessing pipeline approaches using R^2 evaluation scores.

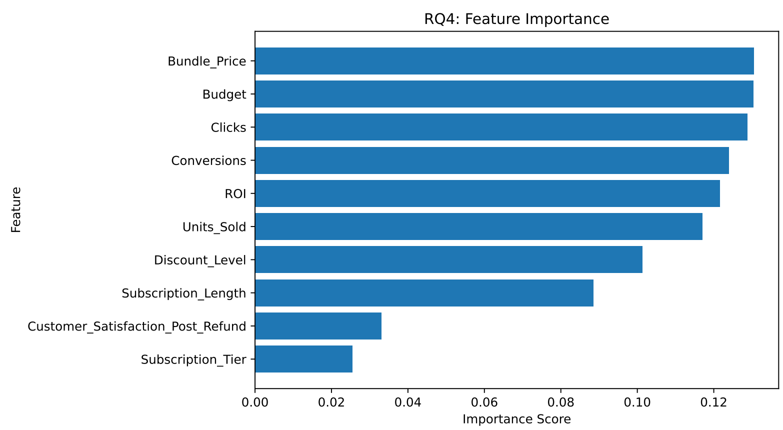

Fig. 4. Feature importance analysis showing the contribution of different marketing and product performance attributes toward revenue prediction.

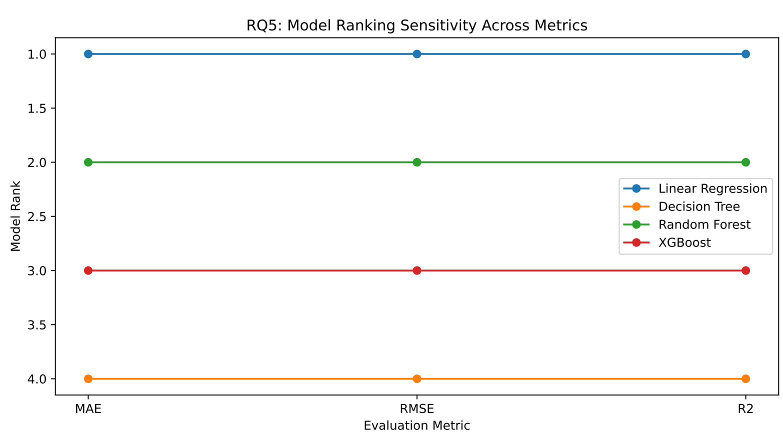

Fig. 5. Model ranking sensitivity analysis across MAE, RMSE, and R^2 evaluation metrics for different regression models.

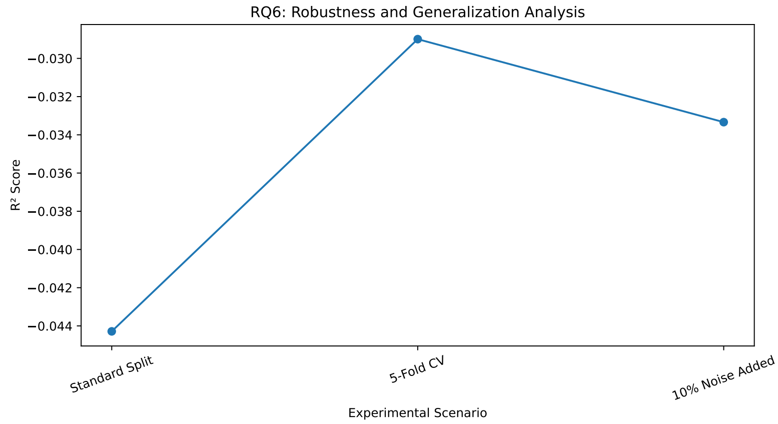

Fig. 6. Robustness and generalization analysis of regression models under different validation scenarios including standard train-test split, cross-validation, and noisy data conditions.

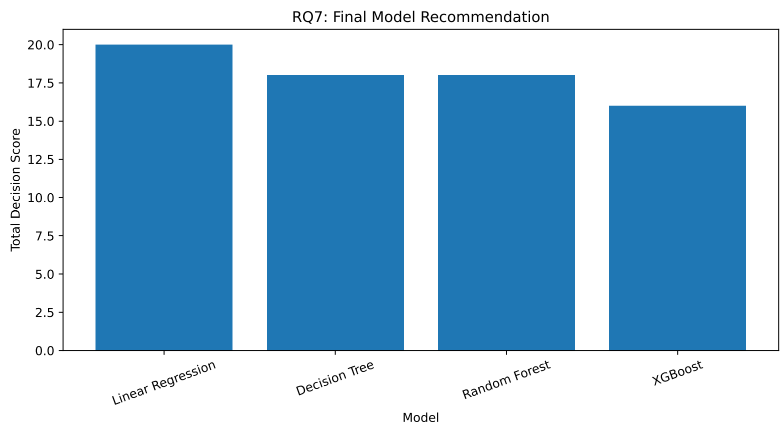

Fig. 7. Final model recommendation analysis based on cumulative decision scores derived from predictive performance, robustness, and evaluation metrics.

IV. Discussion

The experimental analysis demonstrates that supervised learning regression techniques can be effectively applied for marketing revenue prediction using campaign and product performance data. Baseline models such as Linear Regression and K-Nearest Neighbors achieved relatively weak predictive performance because of the complex nonlinear relationships present in the dataset. The Decision Tree model showed improvement after hyperparameter tuning and preprocessing adjustments. However, advanced regression approaches such as SVR, Random Forest, and XGBoost demonstrated improved nonlinear modeling capability and model stability compared to baseline methods.

From the results of the preprocessing tests, feature scaling and comprehensive preprocessing had a positive effect on the regression algorithm. The management of the categorical variables, eliminating unnecessary identification columns, and scaling of numeric features enhanced the performance of the model. The preprocessing impact assessment further emphasized the significance of the preprocessing steps in supervised learning models applied to business analytics.

The result of feature importance analysis was that revenue prediction is highly affected by ROI, Conversions, Units Sold, Budget, and Discount Level because these variables are all related to marketing efforts. This result is in agreement with the practical world of marketing where campaigns are directly responsible for sales which lead to revenues. In the case of robustness analysis, ensemble methods showed better stability and therefore better generalization capability than the baseline regression approach.

On the whole, this research emphasizes the value of using supervised machine learning techniques in marketing analysis and business intelligence tools. This comparison between base model and enhanced model offers helpful information in choosing effective machine learning algorithms for predicting revenue. All these components, including the experiment process, data analysis, feature extraction, and robustness testing, help gain a better understanding of marketing revenue prediction through machine learning models.

A. Future Directions

The future research in this field could be centered around developing more accurate predictions by applying advanced hyperparameter optimization methods, deep learning, and bigger marketing data sets from the real world. Also, other potential fields of study include using time series prediction, clustering analysis, and XAI.

V. Conclusion

In this study, an approach based on supervised learning for predicting the revenue generated by marketing campaigns was proposed. The use of several regression techniques on the Marketing and Product Performance Dataset available at Kaggle included Linear Regression, Decision Tree Regressor, K-Nearest Neighbor Regression, Random Forest, Support Vector Regression, and XGBoost. The results showed that advanced regression models provided broader performance comparison, with SVR producing the strongest R^2 score among the evaluated models. Some preprocessing steps, such as feature scaling, irrelevant attribute removal, and categorical variable encoding, had positive effects on the performance of the algorithms used.

As indicated by the comparative analysis, advanced regression models including SVR, Random Forest, and XGBoost demonstrated stronger nonlinear modeling capability compared to baseline linear approaches. The feature importance analysis also revealed the key features of ROI, Conversions, Budget, and Units Sold contributing to the generation of profit estimates. Robustness analysis performed under varying validation techniques also confirmed that ensemble algorithms exhibited relative robustness in terms of performance.

Generally, it should be noted that the paper presents the applicability of applying supervised learning methods in marketing, business intelligence, and predictive modeling practices. The outlined workflow allows having a systematic methodology for assessing different regression models and identifying their behavior under various conditions related to data preprocessing and model robustness. Further research could involve employing deep learning models and utilizing bigger datasets with advanced optimization techniques.

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